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Day 12

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(38 M)
Q1) Artificial Intelligence (AI) and Precision Farming have been proposed as solutions to improve agricultural productivity in India. Critically analyze their feasibility and challenges in the Indian agricultural landscape.

Ans) Indian agricultural landscape despite its immense potential is marred with various challenges. Given its great agro-ecological diversity, India is recognized as a global agricultural powerhouse. To harness this potential, synergy between AI, precision farming and agricultural productivity is a promising and transformative force.

Artificial Intelligence as solution to improve agricultural productivity

The agricultural landscape in India is undergoing a monumental transformation driven by Artificial Intelligence (AI).

↳ Crop Disease Detection: AI is becoming a game changer in detecting crop diseases. Like neural networks can detect diseases like apple scabs with 95% accuracy. AI based pest surveillance

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Systems are also helping Indian farmers tackle losses created by climate change, ensuring healthy harvests.

↳ Automated Weed Control Systems: Manual weeding or chemical herbicides are both labour-intensive and harmful for the environment. To counter this, Computer vision is used to distinguish weeds from crops and apply herbicides selectively.

↳ Livestock monitoring in real-time: Advanced sensor-based systems and image recognition technologies detect early signs of illnesses, allowing farmers to take prompt action and improve productivity.

↳ AI-Sowing app by Microsoft: The Sowing related text messages give information related to planting times, weed-management, fertilizers application and harvesting.

↳ Price forecasting model: It uses datasets on historical sowing areas, production yields, weather patterns and other relevant information to predict crop yields, crop volumes, commodity price in advance. With this govt. can more accurately plan ahead to set the MSP.

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Precision farming as a solution to improve agricultural productivity.

Precision farming is a combination of different technologies which allows site specific management to efficiently utilize their resources and get economic gains.

- ↳ Satellite Positioning Systems (GPS): A network of orbiting satellites send precise locational details to earth from space to enable farmers to monitor crop conditions.
- ↳ Geographic Information Systems (GIS): It provides information on data parameters used to understand the relationships between factors affecting a crop on a specific site.
- ↳ Sensors (Optical): Provides information on soil properties, plant fertility and water status.
- ↳ Grid Soil Sampling: A method for site specific soil management.
- ↳ Automated irrigation systems: It allow for minimal manual intervention apart from surveillance. Systems like drip, sprinkler or surface irrigation are automated with timers, sensors or computers.

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↳ Laser Land levellers : levelling of a field within a certain degree of the desired slope using a guided laser beam throughout the field.

↳ Variable rate technology : Automates applying materials such as fertilisers, pesticides, seeds and irrigation on the land.

↳ Leaf color chart : It is a ruler-shaped strip with panels of different shades of green. Farmers match the color of their crop leaves to the LCC to find the right amount of fertilizer.

Feasibility of AI and precision farming in the Indian Agricultural landscape

AI & precision farming are transforming Indian agriculture by enabling data-driven decision-making, optimizing resource use and increasing efficiency, leading to sustainable and productive farming practices.

↳ In India, AI and precision farming is yet to become an integral part of mainstream farming systems. It is still in a nascent stage in India.

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- ↳ ICAR has developed a low-cost customized leaf colour chart for nitrogen management in rice which achieved commercial success.
- ↳ This has showcased the scope of low-cost precision farming technologies in India.
- ↳ It is practiced by a limited number of farmers often on a single field or on an experimental basis or commercial farms growing high-value crops.
- ↳ Complete package of these technologies are used only by large progressive farmers as its high cost limits its uptake.
- ↳ With more information on farm performance and needs through these technologies, agricultural loans become more accessible and state governments can support several farming districts at scale.
- ↳ With this farmers can reduce land and water waste while increasing their crop yield.
- ↳ With landscape monitoring models, photos of an individual field's current performance and future needs can be predicted.

Challenges of AI and precision farming in the Indian Agricultural landscape

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- ↳ Challenging for small farmers to utilize the farm machinery in fragmented plots.
- ↳ These technologies are more profitable for high-value crops given the high cost associated with them.
- ↳ Concerns exist regarding the ownership and usage of farm data collected by AI systems as well as the potential for data breaches and misuse.
- ↳ Unreliable power supplies in rural areas can hinder the operation of AI-powered systems and precision farming equipments.
- ↳ Protecting AI systems from cyberattacks is crucial but many farmers lack the technical expertise to implement robust cybersecurity measures.

Given all the facts, there is no doubt that Precision farming's future supported by AI promises to be a revolutionary industry in the Indian agricultural landscape. It prioritizes sustainability and innovation while meeting the needs of an expanding Indian population. But to harness these possibilities, it is important that our farmers become aware and capable to use these technologies. Only then they will be able to take educated decisions which increases their independence and resilience in the face of adversity.